

Early Recall Estimation and Stopping in Technology-Assisted Reviews via Ranker-Sampler Fusion

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Abstract

Technology-Assisted Review (TAR) workflows iteratively prioritize documents for expert screening to achieve high recall (e.g., 0.90–0.95) under limited time and budget. A persistent challenge is deciding when to stop: true recall is unknown during live screening unless all documents are judged, while additional sampling to estimate recall can be prohibitively costly at low prevalence. This paper develops a step-by-step statistical framework to estimate recall early by fusing (i) information from a ranking model and (ii) a small probability sample of documents. We present two complementary estimators: a Bayesian prevalence estimator that treats the ranker as an informative prior updated by a random sample, and a model-assisted estimator (generalized regression estimator with a Horvitz-Thompson residual correction) that combines calibrated ranker probabilities with design-based sampling adjustments. We then derive practical stopping rules based on conservative (lower-bound) recall estimates, and outline a future evaluation plan on established constrained high-recall benchmarks.

CCS Concepts

• **Information systems** → **Information systems applications**; **Retrieval models and ranking**; **Retrieval tasks and goals**; **Evaluation of retrieval results**.

Keywords

Technology-Assisted Review, High-recall retrieval, Stopping rules, Probability sampling, Model-assisted estimation

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1 Introduction

Systematic reviews and evidence syntheses are foundational to evidence-based decision-making [5, 16], yet they remain costly and time-consuming because they require expert screening of large and continuously growing literature. Technology-Assisted Review (TAR) mitigates this burden by iteratively learning from relevance feedback and ranking documents so that relevant items are surfaced

early [12]. In high-recall settings, however, TAR systems are not judged solely by how quickly they retrieve relevant documents, but by whether they can do so under explicit operational constraints: limited reviewer time, limited budgets, and (in many applications) multiple concurrent topics competing for the same pool of human effort. Recent work has shown that ignoring these global constraints can lead to suboptimal or inequitable outcomes, and that budget-aware evaluation and allocation policies can materially affect the number of relevant documents discovered per unit cost [13, 14].

A central unresolved challenge is “stopping”: deciding when the reviewer can terminate screening with confidence that the target recall (typically 0.90 - 0.95) has been achieved. The related work offers a rich set of stopping approaches, including early CAL-based paradigms and knee-style heuristics [6, 8], transparent estimation through adaptive sampling [18], process-based modeling using counting and point processes [22, 23], heuristic procedures tailored to evidence synthesis workflows [3, 25], and more recent reinforcement-learning policies [2]. Despite this progress, most practical workflows still face an uncomfortable trade-off: stopping rules that are conservative enough to protect against missed relevant documents may require substantial additional screening, while aggressive rules can save cost but risk violating the recall target.

This trade-off becomes sharper under global resource constraints. When screening effort must be allocated across multiple topics or review stages, every additional document read for the sake of estimating recall competes directly with reading documents that could yield new relevant evidence. Our prior work has emphasized precisely this allocation problem: Di Nunzio [14] revisited CLEF eHealth TAR through a budget-aware lens, introducing cost-aware metrics and demonstrating that different allocation strategies optimize different notions of efficiency; Di Nunzio [13] further investigated policy optimization under fixed review constraints, showing that topic-sensitive policies can improve yield under limited effort. These findings motivate a stopping methodology that is not merely accurate, but resource-aware: it should achieve reliable recall certification with minimal incremental labeling cost.

In principle, one can estimate recall by taking a random sample from the remaining pool and using observed prevalence to infer how many relevant documents remain. In practice, pure sampling is expensive, particularly when prevalence is low, a common situation in systematic review screening, because tight confidence bounds may require hundreds or thousands of additional judgments. At the same time, TAR systems already produce a rich signal in the form of a learned ranking model. This paper argues that recall estimation should therefore be treated as a fusion problem: combine the statistical validity of a small probability sample with the predictive structure of the ranker so that estimates of total relevant (and hence



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recall) converge earlier, enabling earlier stopping recommendations at a fixed risk tolerance.

In this paper, our focus is on clarifying the theoretical framework and problem setting under study; extensive experimentation and benchmarking are deferred to subsequent work.

We make three contributions:

- First, we formalize stopping under review constraints as estimating total relevant and deriving conservative recall lower bounds that can be used directly in policy decisions about whether to continue screening or reallocate effort.
- Second, we develop a ranker–sampler fusion framework that integrates calibrated ranker probabilities with a small, design-based probability sample to produce (i) a Bayesian prevalence estimator with an interpretable ranker-informed prior and (ii) a model-assisted estimator that corrects ranker bias via sampling residuals. Both estimators are designed to reduce the additional labeling burden relative to sampling-only approaches while preserving defensibility through explicit uncertainty quantification.
- Third, we connect stopping to resource management by analyzing how the marginal cost of additional labeling for recall estimation interacts with budget allocation policies, providing guidance on when it is preferable to stop within a topic versus redirect effort across topics under a fixed global budget [13, 14].

Together, these elements position stopping as an integral component of constrained TAR workflows: not an isolated decision rule, but a statistically grounded mechanism for deciding when continued screening is no longer the best use of limited review resources.

2 Related Work

Stopping strategies for Technology-Assisted Review (TAR) seek to minimize screening workload while achieving high-recall targets, typically 0.90–0.95 [1]

Since true recall is unknown during screening, stopping rules differ primarily in how they (i) infer progress toward the target and (ii) manage the risk of missing relevant documents. From 2020 onward, the literature has consolidated around several complementary families: (a) transparent estimation via residual-relevant or prevalence modeling; (b) heuristic rules designed for operational usability; (c) stochastic-process modeling of relevant-document arrivals; and (d) learning-based stopping policies.

Pioneering work: CAL and knee-style stopping. Much of the modern TAR literature builds on the foundational contributions of Grossman and Cormack, who established continuous active learning (CAL) as a highly effective high-recall review paradigm and provided early, widely reused experimental methodology. Their SIGIR work compared core TAR training protocols (CAL vs. alternatives) using an evaluation toolkit that simulates the reviewer-in-the-loop setting, helping to standardize empirical comparisons in high-recall retrieval [6]. They subsequently advanced autonomous CAL configurations (often referred to as AutoTAR), emphasizing robust performance with minimal topic-specific tuning and a single-seed start-up, which strongly influenced later TAR system designs [7]. On the stopping side, they introduced practical “call-your-shot”

strategies designed to support defensible termination decisions under high-recall objectives, including the widely adopted “knee” criterion based on detecting a negative inflection point in the gain curve (and related reliability-oriented variants) [8, 9]. These contributions effectively set the baseline vocabulary—CAL as the default retrieval policy and knee-style criteria as early operational stopping rules—against which many subsequent estimation-, process-, and learning-based stopping methods are evaluated.

Transparent estimation via sampling and prevalence modeling. A core line of work frames stopping as estimating the total number of relevant documents (or the residual relevant set) so that stopping decisions can be justified in terms of an estimated recall. Li and Kanoulas [18] propose sampling from an adaptive distribution jointly with training a ranking model, proving unbiasedness of their estimators under a with-replacement design and motivating stopping criteria based on estimated residual relevance. More recently, Molinari and Esuli [19] propose SAL τ , a stopping rule that improves prior/prevalence estimates within active learning, showing earlier stopping while retaining target-recall guarantees. Bron et al. [4] introduce a stopping criterion grounded in Chao’s population size estimator, leveraging ideas from population estimation to infer prevalence and guide stopping; they also evaluate an ensemble-based active learning strategy together with the proposed criterion.

Heuristic stopping rules and practitioner-oriented procedures. A second strand emphasizes stopping heuristics that are simple to apply and interpret, often with conservative bias to reduce the risk of erroneous exclusions. Di Nunzio [11] reports a stopping strategy refined through multiple years of participation in the CLEF eHealth TAR task, comparing fixed versus variable resource allocations within a Continuous Active Learning (CAL) framework and analyzing failure modes to identify where stopping heuristics can break down. Yang et al. [25] systematize and evaluate heuristic stopping rules across tasks and recall targets, and propose QUANT and QUANTCI, which use model-based estimation techniques drawn from survey research to better control target recall. In the evidence-synthesis community, Boetjé and van de Schoot [3] propose the SAFE procedure – an diverse and conservative set of heuristics designed for active-learning-based abstract screening (e.g., ASReview), explicitly balancing continued screening effort against the risk of missed relevant studies. Complementing method proposals, König et al. [17] provide a comprehensive empirical evaluation of multiple stopping rules (including knee-based and prevalence-estimation approaches) in AI-aided screening for psychological meta-analytical research, analyzing performance in terms of sensitivity/specificity and screening cost under varying prevalence regimes.

Stochastic-process and event-arrival modeling. Another influential approach models the appearance of relevant documents down the ranking as a counting or point process, producing stopping rules from estimated intensities (rates) of relevant-document arrivals. Sneyd and Stevenson [22] introduce counting-process-based stopping criteria (including Cox process variants and alternative rate functions), demonstrating improvements on CLEF eHealth TAR. Stevenson and Bin-Hezam [23] extend this direction with point-process-based stopping, comparing multiple candidate rate

functions and explicitly evaluating robustness across rankings of varying effectiveness - a key consideration because stopping reliability depends on ranker quality. Bridging process modeling and classifier outputs, Bin-Hezam and Stevenson [2] further explore learning-based stopping (see below), while their process-modeling line of work motivates hybrid designs where model predictions stabilize estimates of remaining relevance.

Learning-based stopping policies. Recent work frames stopping as sequential decision-making. Bin-Hezam and Stevenson [2] propose RLSTOP, a reinforcement-learning-based stopping rule trained on example rankings with a reward function that trades off workload against meeting target recall; experiments across several TAR benchmarks show substantial workload reductions and competitive performance against established baselines. This perspective is attractive for optimizing cost–recall trade-offs, but also raises practical questions about policy transfer and robustness across collections and ranking behaviors, complementing the robustness emphasis found in process-based approaches [23].

Reproducibility and experimental infrastructure. Given the complexity of TAR workflows and the diversity of stopping rules, reproducible experimentation frameworks are increasingly important. Yang and Lewis [24] introduce TAREXP, an open-source Python framework for declarative TAR workflow specification and experimentation, enabling standardized comparisons of ranking, sampling, and stopping components.

Our contribution is intentionally complementary to existing stopping methods rather than a new ranking or active learning policy. In contrast to approaches that tightly couple a specific review policy with a specific estimator (e.g., adaptive sampling intertwined with CAL) [18], we present two modular fusion estimators that can be layered on top of an arbitrary TAR workflow: (i) a Bayesian prevalence estimator in which the ranker induces an informative prior updated by a small probability sample, and (ii) a design-based, model-assisted estimator (GREG with a Horvitz-Thompson residual correction) that uses calibrated document-level probabilities and corrects aggregate model error via probability sampling [10, 15]. Compared to heuristic stopping rules that borrow ideas from survey estimation [25], our estimators explicitly quantify uncertainty (credible/confidence bounds) and separate the ranker-driven judgment stream (used for learning) from a probability sample (used for recall certification). Finally, we frame additional sampling for stopping as an explicit resource trade-off under constrained screening budgets, linking recall certification to effort reallocation decisions across topics [13, 14].

3 Problem Setting

Let $U = \{1, \dots, N\}$ be the corpus of N candidate documents for a systematic review. Each document i has a (latent) relevance label $y_i \in \{0, 1\}$, where $y_i = 1$ indicates relevance to the review. A TAR system maintains a ranker that assigns each document a score $s_i \in \mathbb{R}$ or a predicted relevance probability $\hat{p}_i \in [0, 1]$, and iteratively selects documents for expert assessment.

At iteration t , the expert has judged a set $J_t \subset U$ of size $|J_t| = t$ (or t total judgments), producing labels $\{y_i : i \in J_t\}$. The TAR ranker is updated and the remaining documents $U \setminus J_t$ are re-ranked.

Define:

$$R = \sum_{i \in U} y_i \quad (\text{total number of relevant documents}), \quad (1)$$

$$F_t = \sum_{i \in J_t} y_i \quad (\text{relevant found up to time } t), \quad (2)$$

$$\text{Recall}_t = \frac{F_t}{R}. \quad (3)$$

The key challenge is that R is unknown in live screening, hence Recall_t is unknown.

We seek a stopping strategy that recommends stopping once an estimated recall is sufficiently high:

$$\widehat{\text{Recall}}_t \geq \tau, \quad (4)$$

for a target $\tau \in \{0.90, 0.95\}$, ideally using few additional expert judgments.

A straightforward approach is to take a probability sample $S \subset U$ of size n (e.g., simple random sample without replacement), obtain expert labels on S , and estimate prevalence and thus R .

If $r = \sum_{i \in S} y_i$ and S is an SRSWOR (simple random sample without replacement), an unbiased estimator of prevalence is $\hat{p} = r/n$, and a natural estimator for R is $\hat{R} = N\hat{p}$. However, achieving tight confidence intervals for small prevalence can require large n , which is expensive when each label requires expert screening.

This motivates fusing limited sampling with ranker output so that recall estimation converges earlier.

4 Ranker–Sampler Fusion: Two Estimators

We present two estimators that combine (A) a probability sample (to retain statistical validity) with (B) ranker information (to reduce variance and accelerate convergence).

4.1 Estimator A: Bayesian Prevalence with a Ranker-Informed Prior

This estimator provides a lightweight Bayesian mechanism for combining two complementary sources of information about collection prevalence. The first source is the ranker, which can be interpreted as inducing a prior belief about the proportion of relevant documents in the corpus via either calibrated probabilities or score-based heuristics. The second source is a small probability sample, which supplies design-based evidence that anchors the estimate and mitigates over-reliance on the model. Operationally, we represent prevalence with a Beta prior whose mean is set by the ranker and whose concentration (effective sample size) controls trust in the ranker; we then update this prior with sampled relevance judgments to obtain a posterior prevalence distribution. The resulting posterior yields both a point estimate of total relevant (and hence recall) and a direct route to conservative stopping through posterior credible bounds. We describe this approach through four steps.

Step 1: Summarize the ranker as a prior mean. Assume the ranker provides an estimate of the number of relevant documents, denoted m (with $0 \leq m \leq N$). For example:

$$m = \sum_{i \in U} \hat{p}_i, \quad (5)$$

where $\hat{p}_i$ are calibrated relevance probabilities (see Section 4.2), or m may be a heuristic derived from the score distribution.

Define the implied prevalence:

$$p_0 = \frac{m}{N}. \quad (6)$$

Step 2: Choose a prior strength. We encode the ranker-derived belief as a Beta prior:

$$p \sim \text{Beta}(\alpha_0, \beta_0), \quad (7)$$

with mean $\mathbb{E}[p] = \alpha_0 / (\alpha_0 + \beta_0) = p_0$.

A convenient parameterization uses an effective prior sample size $\kappa > 0$:

$$\alpha_0 = \kappa p_0, \quad \beta_0 = \kappa(1 - p_0). \quad (8)$$

Intuitively, κ controls how strongly we trust the ranker: large κ means the prior dominates; small κ means the data (sample) dominates.

In practice, κ can be set using historical calibration performance, topic similarity, or conservatively chosen (e.g., small-to-moderate values) to avoid over-trusting the ranker.

Step 3: Update with a probability sample. Let S be a probability sample of size n , and let $r = \sum_{i \in S} y_i$ be the number of relevant documents in the sample.

For simplicity, we approximate SRSWOR with a binomial model. Under SRSWOR, the exact sampling distribution of the number of relevant items in the sample is hypergeometric, since draws are dependent due to sampling without replacement: $r \mid R \sim \text{Hypergeometric}(N, R, n)$. When the corpus is large and the sampling fraction is small ($n/N \ll 1$), these dependencies are negligible and the hypergeometric distribution is well-approximated by a binomial model with success probability equal to the population prevalence $p = R/N$ [21, Chapter 12]. Intuitively, each draw changes the remaining prevalence by at most $O(1/N)$, so the sequence of draws behaves almost as if it were independent with replacement. Under this approximation,

$$r \mid p \sim \text{Binomial}(n, p), \quad (9)$$

which yields a conjugate Beta posterior:

$$p \mid r \sim \text{Beta}(\alpha_0 + r, \beta_0 + n - r). \quad (10)$$

Step 4: Estimate total relevant and recall. The posterior mean prevalence is:

$$\hat{p}_{\text{Bayes}} = \mathbb{E}[p \mid r] = \frac{\alpha_0 + r}{\alpha_0 + \beta_0 + n} = \frac{\kappa p_0 + r}{\kappa + n}. \quad (11)$$

Thus,

$$\hat{R}_{\text{Bayes}} = N \hat{p}_{\text{Bayes}}. \quad (12)$$

An estimated recall is then:

$$\widehat{\text{Recall}}_{t, \text{Bayes}} = \frac{F_t}{\hat{R}_{\text{Bayes}}}. \quad (13)$$

As an example of the application of this procedure, if you have sampled n documents and found r relevant, while the ranker suggests m relevant overall, then $p_0 = m/N$ and Eq. (11) yields:

$$\hat{p} = \frac{\kappa(m/N) + r}{\kappa + n}. \quad (14)$$

This is a principled way to combine the sampling evidence (r/n) with the ranker belief (m/N), where κ governs the weight given to the ranker.

Conservative stopping via credible intervals. A stopping rule should be conservative. Let p^U be an upper credible bound for prevalence (e.g., the 95% upper bound from the Beta posterior). Then define:

$$R^U = N p^U, \quad (15)$$

and stop only if a lower bound on recall exceeds the target:

$$\text{LBRecall}_t = \frac{F_t}{R^U} \geq \tau. \quad (16)$$

4.2 Estimator B: Model-Assisted Total-Relevant Estimation (GREG/HT Fusion)

This estimator adopts a model-assisted perspective from survey sampling to fuse ranker outputs with a probability sample at the document level. The key idea is to start from a model-based “soft count” of total relevant documents, $\sum_{i \in U} \hat{p}_i$, where $\hat{p}_i$ is a calibrated probability of relevance produced by the ranker, and then correct any systematic mis-calibration using a design-based sampling adjustment. Concretely, the adjustment is a Horvitz-Thompson (HT) correction [15], which estimates population totals by weighting each sampled unit by the inverse of its inclusion probability π_i ; here it is applied to the residuals ($y_i - \hat{p}_i$) observed in a probability sample. Combining the model-based total with an HT residual correction yields the standard generalized regression (GREG) estimator [10], which is design-consistent under the sampling design and can achieve substantially lower variance when $\hat{p}_i$ is predictive.

The Bayesian estimator above compresses the ranker into a single scalar m . A more informative fusion uses the entire vector of calibrated probabilities $\hat{p}_i$ and corrects it using a probability sample. Once again, we describe the procedure in four steps.

Step 1: Calibrate ranker outputs. Assume the ranker provides a real-valued score s_i . Using judged data (e.g., a held-out subset of J_t or a calibration set), fit a calibration model such as Platt scaling [20]:

$$\hat{p}_i = \sigma(as_i + b) = \frac{1}{1 + \exp(-(as_i + b))}. \quad (17)$$

Calibration makes $\hat{p}_i$ interpretable as an estimated probability of relevance.

Step 2: Draw a probability sample and compute inclusion probabilities. Let S be a probability sample with known inclusion probabilities $\pi_i = \Pr(i \in S)$. Examples include:

- SRSWOR: $\pi_i = n/N$ for all i .
- Stratified sampling over rank buckets (e.g., top- k bands) with known π_i per stratum.
- Unequal-probability sampling with known π_i under the design.

These inclusion probabilities define Horvitz-Thompson weights $w_i = 1/\pi_i$, which are used in Step 3 to estimate the population total of residuals $e_i = y_i - \hat{p}_i$ from the probability sample.

Step 3: Model-assisted estimator for total relevant. A model-assisted (generalized regression) estimator for the total relevant count $R =$

$\sum_{i \in U} y_i$ is:

$$\hat{R}_{MA} = \sum_{i \in U} \hat{p}_i + \sum_{i \in S} \frac{y_i - \hat{p}_i}{\pi_i}. \quad (18)$$

This is a standard model-assisted (GREG) estimator: it is design-consistent under the sampling design and gains efficiency when $\hat{p}_i$ is predictive. The first term is the ranker-implied total relevant (soft count). The second term is a Horvitz–Thompson correction that estimates the total residual relevance and adjusts the model-based total accordingly.

Under simple random sampling without replacement (SRSWOR), $\pi_i = n/N$ for all sampled documents, so the Horvitz–Thompson correction in Eq. (18) simplifies to a scaled sample mean:

$$\hat{R}_{MA} = \sum_{i \in U} \hat{p}_i + \frac{N}{n} \sum_{i \in S} (y_i - \hat{p}_i) = \sum_{i \in U} \hat{p}_i + N \bar{e}_S, \quad e_i = y_i - \hat{p}_i. \quad (19)$$

This form is useful for interpretation. The first term is the ranker-implied “soft” total relevant, while the second term adds (or subtracts) an estimate of the ranker’s *average aggregate error* based on the probability sample: $\bar{e}_S$ is the mean residual (miscalibration) on the sample, and $N \bar{e}_S$ extrapolates that mean error to the full corpus. Consequently, if the ranker is well calibrated on average, $\bar{e}_S \approx 0$ and the correction is small; if the ranker systematically underestimates prevalence then $\bar{e}_S > 0$ and $\hat{R}_{MA}$ increases; if it overestimates prevalence then $\bar{e}_S < 0$ and $\hat{R}_{MA}$ decreases. This also clarifies why ranker–sampler fusion can converge with relatively small n : when $\hat{p}_i$ captures most of the variation in relevance, the residuals have lower variance than the raw labels, so estimating $\bar{e}_S$ requires fewer additional judgments than estimating prevalence directly from y_i alone.

Step 4: Recall estimate and conservative stopping. Estimate recall at time t as:

$$\widehat{\text{Recall}}_{t,MA} = \frac{F_t}{\hat{R}_{MA}}. \quad (20)$$

For stopping, compute an upper confidence bound for R (or equivalently a lower confidence bound for recall). A conservative option is:

$$R^U = \hat{R}_{MA} + z_{1-\alpha} \sqrt{\widehat{\text{Var}}(\hat{R}_{MA})}, \quad (21)$$

then stop if:

$$\text{LBRecall}_t = \frac{F_t}{R^U} \geq \tau. \quad (22)$$

Variance estimation depends on the sampling design; for SRSWOR, a standard approximation is:

$$\widehat{\text{Var}}(\hat{R}_{MA}) \approx N^2 \left(1 - \frac{n}{N}\right) \frac{s_e^2}{n}, \quad s_e^2 = \frac{1}{n-1} \sum_{i \in S} (e_i - \bar{e})^2, \quad e_i = y_i - \hat{p}_i. \quad (23)$$

4.3 Stopping Procedure

Algorithm 1 summarizes a practical implementation that (i) runs normal TAR screening, (ii) intermittently draws a small probability sample (possibly stratified by rank), (iii) updates the recall estimate, and (iv) stops when a conservative recall bound exceeds τ .

Algorithm 1 Ranker–Sampler Fusion for Recall Estimation and Stopping

Require: Corpus U with $|U| = N$, target recall τ , significance α

- 1: Initialize ranker and empty judged set $J_0 \leftarrow \emptyset$, found relevant $F_0 \leftarrow 0$
- 2: **for** $t = 1, 2, \dots$ **do**
- 3: Select next document $i_t \in U \setminus J_{t-1}$ (e.g., top-ranked)
- 4: Obtain expert label $y_{i_t} \in \{0, 1\}$
- 5: Update $J_t \leftarrow J_{t-1} \cup \{i_t\}$; update $F_t \leftarrow F_{t-1} + y_{i_t}$
- 6: Update ranker using $(J_t, \{y_i\}_{i \in J_t})$; compute scores s_i for $i \in U$
- 7: Calibrate to probabilities $\hat{p}_i$ using judged data
- 8: **if** stopping check scheduled (e.g., every B judgments) **then**
- 9: Draw probability sample S (size n) from $U \setminus J_t$ with known π_i
- 10: Expert-label sampled items to obtain $\{y_i\}_{i \in S}$
- 11: Compute $\hat{R}$ using Bayesian estimator (Eq. (11)) or model-assisted estimator (Eq. (18))
- 12: Compute conservative upper bound R^U (credible or confidence bound)
- 13: **if** $F_t/R^U \geq \tau$ **then**
- 14: **return** Stop and report estimated recall $\geq \tau$
- 15: **end if**
- 16: **end if**
- 17: **end for**

5 Simulation Study

To support an interpretable, end-to-end validation of recall estimation and stopping under controlled conditions, we conduct a simulation study in which the *ground truth* relevance labels are known. This enables computation of the *oracle recall* at each screening depth and direct comparison against the proposed estimators and their uncertainty quantification.

5.1 Synthetic corpus generation

We generate a corpus $U = \{1, \dots, N\}$ of size N with latent relevance labels

$$y_i \sim \text{Bernoulli}(p), \quad i = 1, \dots, N, \quad (24)$$

where p controls prevalence. In our default setting we use $N = 10,000$ and $p = 0.10$.

To simulate a ranker with controllable discrimination, we generate continuous scores s_i from a two-component Gaussian model:

$$s_i \mid y_i = 1 \sim \mathcal{N}(\mu_1, \sigma^2), \quad s_i \mid y_i = 0 \sim \mathcal{N}(\mu_0, \sigma^2), \quad (25)$$

where $\Delta = \mu_1 - \mu_0$ determines separability. For equal-variance Gaussians, the area under the ROC curve satisfies

$$\text{AUC} = \Phi\left(\frac{\Delta}{\sqrt{2}\sigma}\right), \quad (26)$$

so we set $\Delta = \sqrt{2}\sigma \Phi^{-1}(\text{AUC})$ to achieve a target AUC (default: AUC = 0.80). Documents are ranked by decreasing score, yielding a static screening order ord.

Finally, we derive ranker probabilities by mapping scores to the unit interval via a logistic transform,

$$\hat{p}_i = \sigma(\beta s_i + b), \quad (27)$$

with optional perturbations on the logit scale to model miscalibration (e.g., logit shift and/or slope distortion). This allows stress-testing of the fusion estimators under controlled calibration error.

5.2 TAR screening trajectory and oracle recall

Given the ranking ord, we simulate screening by judging the top- t documents,

$$J_t = \{\text{ord}_1, \dots, \text{ord}_t\}, \quad F_t = \sum_{i \in J_t} y_i. \quad (28)$$

Because the simulation reveals all labels, we can compute the total number of relevant documents

$$R_{\text{true}} = \sum_{i \in U} y_i, \quad (29)$$

and thus the *oracle recall* at screening depth t :

$$\text{Recall}_t^* = \frac{F_t}{R_{\text{true}}}. \quad (30)$$

In real TAR deployments R_{true} is unknown, so Recall_t^* serves purely as a ground-truth benchmark for evaluating estimator accuracy and uncertainty.

5.3 Probability sampling for recall estimation

At a set of checkpoint depths $t \in \mathcal{T}$ (e.g., every B judgments), we draw a probability sample S_t from the *remaining* unjudged pool $U \setminus J_t$. In the default configuration we use SRSWOR with sampling fraction f :

$$|S_t| = n = \lfloor fN \rfloor, \quad \pi_i = \Pr(i \in S_t) = \frac{n}{|U \setminus J_t|}, \quad (31)$$

with $f = 0.05$ (so $n \approx 500$ when $N = 10,000$). Sampled documents are labeled using the known y_i to emulate expert assessment.

5.4 Estimators, uncertainty, and stopping

At each checkpoint $t \in \mathcal{T}$ we compute two recall estimators on the remaining pool, yielding an estimated total relevant $\hat{R}_t$ and a conservative upper bound R_t^U , which in turn define a lower bound on recall,

$$\text{LBRecall}_t = \frac{F_t}{R_t^U}. \quad (32)$$

We declare stopping when $\text{LBRecall}_t \geq \tau$ for a target τ (default: $\tau = 0.90$).

Bayesian estimator. We summarize the ranker into a prior mean prevalence over the remaining pool via $p_0 = \sum_{i \in U \setminus J_t} \hat{p}_i / |U \setminus J_t|$, set a Beta prior with effective strength κ , and update with sample outcomes using a binomial approximation to SRSWOR. Uncertainty is quantified using posterior credible intervals, and R_t^U is formed from an upper posterior quantile of prevalence.

Model-assisted (GREG/HT) estimator. We form a model-assisted estimate of the remaining total relevant as the sum of predicted probabilities plus a Horvitz–Thompson correction on sampled residuals $(y_i - \hat{p}_i)$. Uncertainty is quantified with a design-based variance approximation computed from sampled residuals, yielding an upper confidence bound R_t^U .

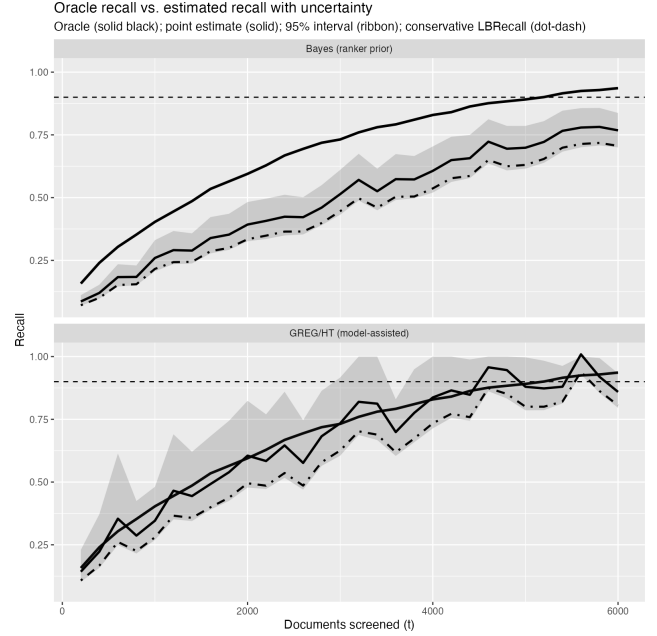


Figure 1: Oracle recall vs. estimated recall with uncertainty along a synthetic TAR screening trajectory ($N=10,000$, prevalence $p=0.05$, ranker $\text{AUC}=0.80$; probability sample fraction $f=0.05$ drawn from the remaining pool at each checkpoint; target $\tau=0.90$). The thick black curve is the oracle recall F_t/R_{true} (available only in simulation). In each panel, the solid curve is the point estimate of recall for the corresponding estimator (Bayes ranker-informed prior, top; GREG/HT model-assisted, bottom), the shaded band is a 95% uncertainty interval, and the dot-dash curve is the conservative lower-bound recall $\text{LBRecall}_t = F_t/R_t^U$ used for stopping. The horizontal dashed line marks the target recall τ .

5.5 Evaluation protocol and visual diagnostics

We compare estimators against the oracle along the screening trajectory. For each $t \in \mathcal{T}$ we report: (i) oracle recall Recall_t^* ; (ii) point estimates $\hat{\text{Recall}}_t$; (iii) 95% uncertainty intervals for $\hat{\text{Recall}}_t$; and (iv) the conservative lower bound LBRecall_t used for stopping. We also compute the *stopping time* for each method as the smallest t at which $\text{LBRecall}_t \geq \tau$, and compare it with the oracle time at which $\text{Recall}_t^* \geq \tau$.

To summarize variability, we repeat the simulation across multiple random seeds, regenerating labels and scores each time while holding $(N, p, \text{AUC}, f, \tau, \kappa)$ fixed. This allows estimation of the distribution of stopping times, false-stop frequency (stopping while $\text{Recall}_t^* < \tau$), and empirical coverage of R_t^U (i.e., $\Pr(R_{\text{true}} \leq R_t^U)$).

We simulated a corpus of size $N=10,000$ with low prevalence ($p=0.05$) and a moderately strong ranker (target $\text{AUC}=0.80$), and tracked recall as documents were screened in decreasing score order. The solid black curve is the *oracle recall* $\text{Recall}_t^* = F_t/R_{\text{true}}$, which is only available in simulation because R_{true} is known. At each checkpoint t , we drew a probability sample from the remaining pool with sampling fraction $f=0.05$ (about 500 documents when

the remaining pool is near N) and computed (i) a Bayesian ranker-informed prevalence update and (ii) a GREG/HT model-assisted correction using sampled residuals.

The result is plotted in Figure 1. The plots show that the Bayesian estimator substantially underestimates recall throughout the trajectory: its point estimate and conservative lower bound lie well below the oracle curve. This behavior is consistent with a ranker prior that implies an overly large total-relevant count (e.g., $\sum_i \hat{p}_i > R_{\text{true}}$ due to imperfect calibration), which inflates $\hat{R}$ and thus depresses $\text{Recall}_t = F_t/\hat{R}$. In contrast, the GREG/HT estimator tracks the oracle recall more closely, reflecting the fact that the Horvitz-Thompson residual term uses the probability sample to correct the ranker's aggregate calibration error. However, uncertainty remains wide, especially early in the process, because low prevalence yields few positive observations in the probability sample, increasing residual variability and producing broader confidence bands (and hence more conservative LBRecall_t). Overall, this run illustrates the intended trade-off: when prevalence is low, fusion is valuable, but performance hinges on probability calibration (for the Bayesian prior) and on sufficient sampling signal to stabilize the residual correction and its uncertainty (for GREG/HT).

6 Discussion and Limitations

Our ranker-sampler fusion perspective is most beneficial in settings where (i) the ranker is reasonably calibrated (or can be calibrated online), (ii) the prevalence of relevant documents is low, a common characteristic of systematic review screening, and (iii) a modest probability sample is feasible but a large sample is prohibitive. In such regimes, using the ranker to explain much of the relevance structure and using a small probability sample primarily to correct residual error can yield earlier and more stable estimates of total relevant documents, and thus more actionable recall-based stopping recommendations. Overall, these considerations highlight that effective stopping under constraints is not solely a matter of ranking quality: it depends on careful integration of ranker outputs with statistically principled sampling designs and conservative uncertainty handling. In particular, low prevalence makes unbiased sampling expensive, while the ranker provides structure that turns prevalence estimation into estimating a small correction term, which typically converges faster. In follow-on work, we will evaluate these estimators and stopping bounds under realistic screening budgets and topic-allocation policies to quantify the practical gains in workload reduction and recall reliability.

The two estimators presented offer complementary trade-offs. The Bayesian estimator is lightweight and interpretable: it summarizes ranker information into a prior mean and uses an effective prior sample size κ to control trust in the ranker. Its simplicity makes it attractive for deployment, but performance depends on selecting κ in a way that reflects calibration reliability and domain shift. By contrast, the model-assisted (GREG) estimator exploits document-level probabilities $\hat{p}_i$ and uses a Horvitz-Thompson correction on sampled residuals, which can substantially reduce variance when $\hat{p}_i$ tracks y_i well. This estimator is design-consistent under the sampling plan and is therefore well suited for settings that require defensible inference, provided that inclusion probabilities are correctly specified.

Several limitations motivate practical safeguards. First, both estimators can fail if the ranker is systematically biased and the probability-sampling component is too small to detect and correct the bias. Conservative configurations include using small κ (so that sampled evidence dominates the Bayesian update), employing stratified sampling that guarantees coverage of lower-ranked regions where residual relevant documents may concentrate, and making stopping decisions based on a lower bound on recall (e.g., LBRecall_t) rather than a point estimate. Second, the primary TAR judgment stream is inherently selection-biased because it follows the ranker and is not a probability sample; it should therefore be treated as training data for ranking and calibration, while uncertainty quantification and recall certification should rely on a separate probability sample drawn under a known design. Finally, when the sampling fraction is non-negligible or the corpus is small, finite-population effects may require replacing binomial approximations with hypergeometric likelihoods or incorporating finite-population corrections, at the cost of additional modeling complexity.

We also plan to evaluate the proposed stopping estimators in constrained, benchmarked high-recall settings that reflect realistic review conditions. In particular, we will use collections and protocols from the CLEF eHealth Technology-Assisted Review tasks and the TREC Total Recall and Legal tracks, which provide established corpora, topic sets, and evaluation methodologies for studying high-recall screening under limited effort. These testbeds will allow us to (i) compare against widely used stopping baselines and recent state-of-the-art methods, (ii) stress-test robustness across rankings of varying effectiveness, and (iii) quantify how much additional sampling overhead is required to achieve conservative recall certification at targets such as 0.90 and 0.95.

References

- [1] Rebecca Beynon, Mariska M. G. Leeflang, Steve McDonald, Anne Eisinga, Ruth L. Mitchell, Penny Whiting, and Julie M. Glanville. 2013. Search Strategies to Identify Diagnostic Accuracy Studies in MEDLINE and EMBASE. *The Cochrane Database of Systematic Reviews* 2013, 9 (Sept. 2013), MR000022. doi:10.1002/14651858.MR000022.pub3
- [2] Reem Bin-Hezam and Mark Stevenson. 2024. RLStop: A Reinforcement Learning Stopping Method for TAR. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*. Association for Computing Machinery, New York, NY, USA, 2604–2608. doi:10.1145/3626772.3657911
- [3] Josien Boetjé and Rens van de Schoot. 2024. The SAFE Procedure: A Practical Stopping Heuristic for Active Learning-Based Screening in Systematic Reviews and Meta-Analyses. *Systematic Reviews* 13, 1 (March 2024), 81. doi:10.1186/s13643-024-02502-7
- [4] Michiel Bron, Peter G. M. van der Heijden, Ad Feelders, and Arno Siebes. 2025. Using Chao's Estimator as a Stopping Criterion for Technology-Assisted Review. *ACM Trans. Inf. Syst.* 43, 3 (May 2025), 81:1–81:51. doi:10.1145/3724116
- [5] Deborah J. Cook, Cynthia D. Mulrow, and R. Brian Haynes. 1997. Systematic Reviews: Synthesis of Best Evidence for Clinical Decisions. *Annals of Internal Medicine* 126, 5 (March 1997), 376–380. doi:10.7326/0003-4819-126-5-199703010-00006
- [6] Gordon V. Cormack and Maura R. Grossman. 2014. Evaluation of Machine-Learning Protocols for Technology-Assisted Review in Electronic Discovery. In *Proceedings of the 37th International ACM SIGIR Conference on Research & Development in Information Retrieval (SIGIR '14)*. Association for Computing Machinery, New York, NY, USA, 153–162. doi:10.1145/2600428.2609601
- [7] Gordon V. Cormack and Maura R. Grossman. 2015. Autonomy and Reliability of Continuous Active Learning for Technology-Assisted Review. arXiv:1504.06868 [cs] doi:10.48550/arXiv.1504.06868
- [8] Gordon V. Cormack and Maura R. Grossman. 2016. Engineering Quality and Reliability in Technology-Assisted Review. In *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '16)*. Association for Computing Machinery, New York, NY, USA, 75–84. doi:10.1145/2911451.2911510

- [9] Gordon V. Cormack and Maura R. Grossman. 2016. "When to Stop" Waterloo (Cormack) Participation in the TREC 2016 Total Recall Track. In *Proceedings of The Twenty-Fifth Text REtrieval Conference, TREC 2016, Gaithersburg, Maryland, USA, November 15–18, 2016 (NIST Special Publication, Vol. 500-321)*, Ellen M. Voorhees and Angela Ellis (Eds.). National Institute of Standards and Technology (NIST). <http://trec.nist.gov/pubs/trec25/papers/WaterlooCormack-TR.pdf>
- [10] Jean-Claude Deville and Carl-Erik Särndal. 1992. Calibration Estimators in Survey Sampling. *J. Amer. Statist. Assoc.* 87, 418 (June 1992), 376–382. doi:10.1080/01621459.1992.10475217
- [11] Giorgio Maria Di Nunzio. 2020. A Study on a Stopping Strategy for Systematic Reviews Based on a Distributed Effort Approach. In *Experimental IR Meets Multilinguality, Multimodality, and Interaction*, Avi Arampatzis, Evangelos Kanoulas, Theodora Tsikrika, Stefanos Vrochidis, Hideo Joho, Christina Lioma, Carsten Eickhoff, Aurélie Névél, Linda Cappellato, and Nicola Ferro (Eds.). Springer International Publishing, Cham, 112–123. doi:10.1007/978-3-030-58219-7_10
- [12] Giorgio Maria Di Nunzio. 2024. Technology Assisted Review Systems: Current and Future Directions. In *Proceedings of the 3rd Workshop on Augmented Intelligence for Technology-Assisted Reviews Systems (ALTARS 2024) (CEUR Workshop Proceedings, Vol. 3832)*, Giorgio Maria Di Nunzio, Evangelos Kanoulas, and Prasenjit Majumder (Eds.). CEUR, Glasgow, Scotland.
- [13] Giorgio Maria Di Nunzio. 2025. POLAR: Policy Optimization for Literature Analysis under Review Constraints. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25)*. Association for Computing Machinery, New York, NY, USA, 4712–4716. doi:10.1145/3746252.3760834
- [14] Giorgio Maria Di Nunzio. 2025. Screening Smarter, Not Harder: Budget Allocation Strategies for Technology-Assisted Reviews (TARs) in Empirical Medicine. *Machine Learning and Knowledge Extraction* 7, 3 (Sept. 2025), 104. doi:10.3390/make7030104
- [15] D. G. Horvitz and D. J. Thompson. 1952. A Generalization of Sampling Without Replacement From a Finite Universe. *J. Amer. Statist. Assoc.* 47, 260 (1952), 663–685. jstor:2280784 doi:10.2307/2280784
- [16] Institute of Medicine (US) Committee on Standards for Systematic Reviews of Comparative Effectiveness Research. 2011. *Finding What Works in Health Care: Standards for Systematic Reviews*. National Academies Press (US), Washington (DC).
- [17] Lars König, Steffen Zitzmann, Tim Fütterer, Diego G. Campos, Ronny Scherer, and Martin Hecht. 2024. An Evaluation of the Performance of Stopping Rules in AI-aided Screening for Psychological Meta-Analytical Research. *Research Synthesis Methods* 15, 6 (2024), 1120–1146. doi:10.1002/jrsm.1762
- [18] Dan Li and Evangelos Kanoulas. 2020. When to Stop Reviewing in Technology-Assisted Reviews: Sampling from an Adaptive Distribution to Estimate Residual Relevant Documents. *ACM Trans. Inf. Syst.* 38, 4 (Sept. 2020), 41:1–41:36. doi:10.1145/3411755
- [19] Alessio Molinari and Andrea Esuli. 2024. SALr: Efficiently Stopping TAR by Improving Priors Estimates. *Data Mining and Knowledge Discovery* 38, 2 (March 2024), 535–568. doi:10.1007/s10618-023-00961-5
- [20] Alexandru Niculescu-Mizil and Rich Caruana. 2005. Predicting Good Probabilities with Supervised Learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML '05)*. Association for Computing Machinery, New York, NY, USA, 625–632. doi:10.1145/1102351.1102430
- [21] Kyle Siegrist. 2022. The Hypergeometric Distribution: Convergence of the Hypergeometric Distribution to the Binomial. [https://stats.libretexts.org/Bookshelves/Probability_Theory/Probability_Mathematical_Statistics_and_Stochastic_Processes_\(Siegrist\)/Statistics_LibreTexts](https://stats.libretexts.org/Bookshelves/Probability_Theory/Probability_Mathematical_Statistics_and_Stochastic_Processes_(Siegrist)/Statistics_LibreTexts), accessed 2026-01-26.
- [22] Alison Sneyd and Mark Stevenson. 2021. Stopping Criteria for Technology Assisted Reviews Based on Counting Processes. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '21)*. Association for Computing Machinery, New York, NY, USA, 2293–2297. doi:10.1145/3404835.3463013
- [23] Mark Stevenson and Reem Bin-Hezam. 2023. Stopping Methods for Technology-assisted Reviews Based on Point Processes. *ACM Trans. Inf. Syst.* 42, 3 (Dec. 2023), 73:1–73:37. doi:10.1145/3631990
- [24] Eugene Yang and David D. Lewis. 2022. TARexp: A Python Framework for Technology-Assisted Review Experiments. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '22)*. Association for Computing Machinery, New York, NY, USA, 3256–3261. doi:10.1145/3477495.3531663
- [25] Eugene Yang, David D. Lewis, and Ophir Frieder. 2021. Heuristic Stopping Rules for Technology-Assisted Review. In *Proceedings of the 21st ACM Symposium on Document Engineering (DocEng '21)*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/3469096.3469873